Abdelrahman Khalifa

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Summary

Software engineer in modern C++, Rust, C and Python, strong in algorithms (Codeforces Expert). Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting): built the Printers page for COSMIC Settings, with 11 PRs merged upstream.

Experience

Google Summer of Code Contributor May 2026 – Oct 2026
The Linux Foundation (OpenPrinting) *Remote*

- Built the **Printers page for COSMIC Settings** and a standalone app for printer status, supply levels, print queues and adding printers (cosmic-settings #2182, in review).
- Designed it as 5 Rust crates and ran the C FFI backend as a varlink IPC service in the settings daemon, so a native crash cannot take down Settings.
- **11 PRs merged upstream** (list): ported cups-rs to libcups3 with CI, added DNS-SD bindings, and fixed crashes and a leaked client in libcups (C).

Embedded Software Engineer Aug 2025 – Present
EKSON *Egypt*

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS: PID valve control with IMU feedback, safety stops and a UART protocol.
- Cut CPU use and latency of the PyQt5 control app by about 40% (measured) using asyncio.
- Engineered remote operation and firmware updates: a Node.js/Express + MongoDB backend polled over REST, with esptool flashing; GitHub Actions CI.

Embedded Linux Systems Intern Dec 2024 – Jan 2025
Orange Digital Center (ODC) *Egypt*

- Built custom Linux images with Yocto and Buildroot and a Linux kernel module; scored 100% in the final evaluation.

OSS Projects

Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | *Contributor (all merged)* Mar 2025 – Dec 2025

- Zephyr RTOS: added the `IP_MULTICAST_LOOP` socket option for IPv4 in 6 networking modules (#86977).
- Boa: optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion (PR #4553).
- Rust Clippy: fixed `ref_as_ptr` false positives for non-temporary references in `let/const` (PR #16214).

Projects

Database Systems, CMU 15-445 | *C++17* GitHub

- Thread-safe Count-Min Sketch for BusTub (per-row locks, merge, top-k); SQL in DuckDB.

Kernel Monitor | *C, Linux, ARM* GitHub

- Character device driver exporting per-task memory, threads and CPU time (RCU).

Technical Skills

Languages: Rust, Modern C++, C, Python (asyncio), JavaScript (Node.js), SQL, Java, C#, Bash
Computer science: Data structures and algorithms, operating systems, concurrency, memory management, networking (TCP/UDP)
Backend & tools: Node.js, Express, REST APIs, MongoDB (NoSQL), PyQt5; Git, GitHub Actions CI, CMake, testing, debugging, code review

Education

Minia University Minia, Egypt
Bachelor of Science (B.Sc.), Computer and Systems Engineering *2020 – 2025*

- Grade: Very Good with Honors.
- Codeforces Expert (handle _Khalifa); 4-time ECPC finalist, 2021–2024.
- Treasurer, IEEE Student Branch, Minia University: managed a budget of more than $10k.